Introduction to Data Science

Unit 1

- **Data science:** Basic terminology - Example – Sigma Technologies -
- **Types of Data:** Structured versus unstructured data - Quantitative versus qualitative data - The four levels of data – Nominal – Ordinal – Interval – Ratio

Introduction

- 19th century: *industrial age*. Mankind was exploring its place in industry alongside giant mechanical inventions.
- 20th century: industrial age was over and was replaced by *information age*. We started using machines to gather and store information (data) about ourselves and our environment for the purpose of understanding our universe.
- 21st century: a problem - we have so much data and we keep making more. Data scientist - using data to make accurate predictions and simulations

What is data science?

- **Data**: collection of information in either an *organized* or *unorganized* format
  - **Organized data**: This refers to data that is sorted into a row/column structure, where every row represents a single *observation* and the columns represent the *characteristics* of that observation.
  - **Unorganized data**: This is the type of data that is in the free form, usually text or raw audio/signals that must be parsed further to become organized.
- **Data Science**: is the art and science of acquiring knowledge through data. It is all about how we take data, use it to acquire knowledge, and then use that knowledge to do the following:
    - Make decisions
    - Predict the future
    - Understand the past/present
    - Create new industries/products

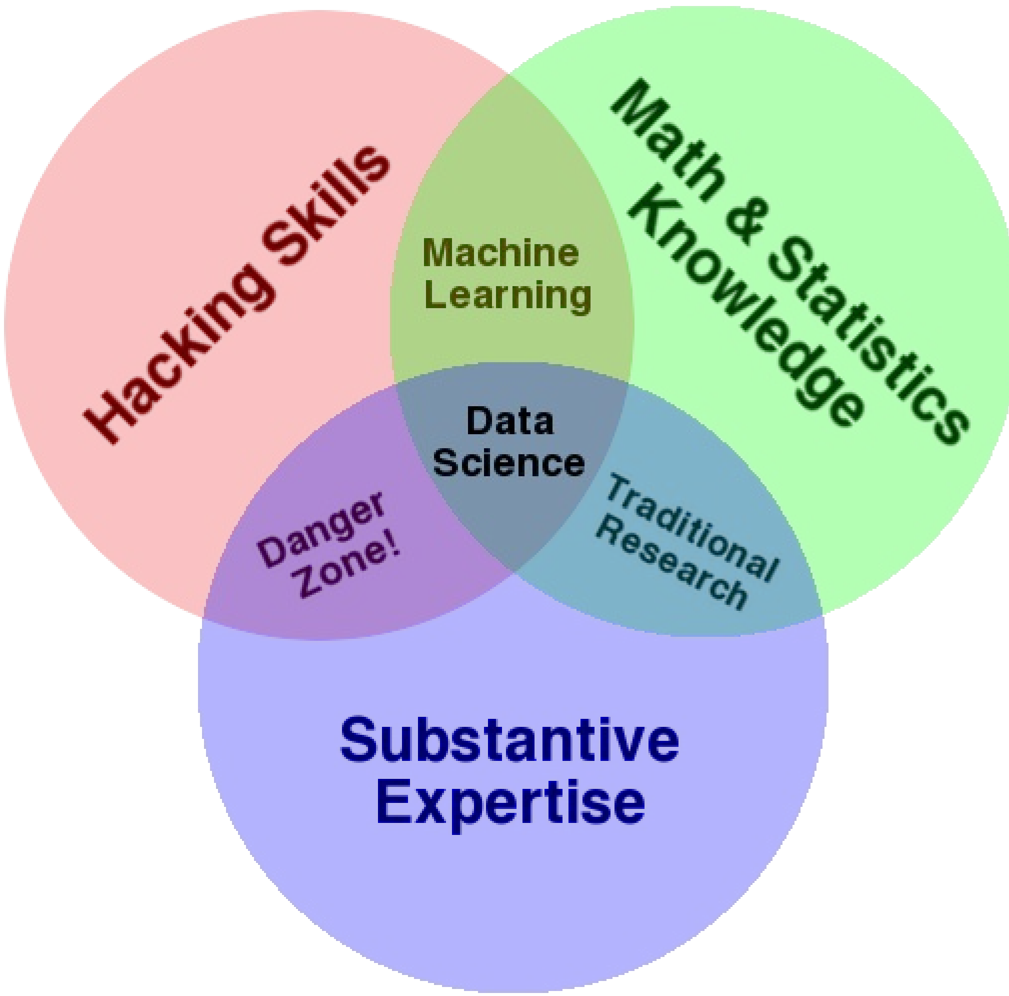
Data Science

Understanding data science begins with three basic areas:

- **Math/statistics**: use of equations and formulas to perform analysis
- **Computer programming**: ability to use code to create outcomes on the computer
- **Domain knowledge**: This refers to understanding the problem domain (medicine, finance, social science, and so on)

- Having a **Math & Statistics Knowledge** base allows you to theorize and evaluate algorithms and tweak the existing procedures to fit specific situations.
- Having **Substantive Expertise** (domain expertise) allows you to apply concepts and results in a meaningful and effective way.
- In order to gain knowledge from data, we must be able to utilize computer programming to access the data, understand the mathematics behind the models we derive, and above all, understand our analyses' place in the domain we are in.

Venn Diagram of Data Science



Why Data Science?

- Data is collected in various forms and from different sources, and often comes in very unorganized.
- The sheer volume of data makes it literally impossible for a human to parse it in a reasonable time.
- Data can be missing, incomplete, or just flat out wrong.
- Often, we have data on very different scales and that makes it tough to compare it.
- One of the main goals of data science: to make explicit practices and procedures to discover and apply the relationships in the data.

Example – Sigma Technologies

- Ben Runkle, CEO, Sigma Technologies
- His chief data scientist, Dr. Jessie Hughan
- A huge problem -The company is consistently losing long-time customers.
- CEO does not know why they are leaving, but he feels he must do something fast. Data Scientist turns to the transcripts of recent customer service tickets
- CEO is convinced that in order to reduce his churn, he must create new products and features, and consolidate existing technologies.
- It is clear that customers were having problems with the existing UI/UX, and weren't upset due to a lack of features. Runkle and Hughan organized a mass UI/UX overhaul and their sales have never been better.

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- Today's common stick-to-your-gut CEO wants to make all decisions quickly and iterate over solutions until something works.
- Dr. Haghun is much more analytical. She wants to solve the problem just as much as Runkle, but she turns to user-generated data instead of her gut feeling for answers.
- Data science is about applying the skills of the analytical mind and using them as a driver would.
- Hagun's way of thinking that dominates the ideas of data science—using data generated by the company as her source of information rather than just picking up a solution and going with it.

Types of Data
Structured versus unstructured data:

- **Structured (organized) data**: This is data that can be thought of as observations and characteristics. It is usually organized using a table method (rows and columns).
- **Unstructured (unorganized) data**: This data exists as a free entity and does not follow any standard organization hierarchy.

• Examples:
  - Most data that exists in text form, including server logs and Face book posts, is *unstructured*
  - Scientific observations, as recorded by careful scientists, are kept in a very neat and organized (*structured*) format

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- A genetic sequence of chemical nucleotides (for example, ACGTATTGCA) is *unstructured* even if the order of the nucleotides matters as we cannot form descriptors of the sequence using a row/column format without taking a further look

- Structured data is generally thought of as being much easier to work with and analyze. Most statistical and machine learning models were built with structured data in mind and cannot work on the loose interpretation of unstructured data. So, with most of our data existing in this free-form format, we must turn to pre analysis techniques, called *preprocessing*, in order to apply structure to at least a part of the data for further analysis

Example for Preprocessing

- *This Wednesday morn, are you early to rise? Then look East. The Crescent Moon joins Venus & Saturn. Afloat in the dawn skies.*
- Pre-processing allows us to explore features that have been created from the existing features. For example, we can extract features such as word count and special characters from the mentioned tweet. Now, let's take a look at a few features that we can extract from text.
- Word/phrase count

	this	wednesday	morn	are	this wednesday
Word Count	1	1	1	1	1

- Presence of certain special characters

	this	wednesday	morn	are	this wednesday	?
Word Count	1	1	1	1	1	1

- **Relative length of the tweet:** The average tweet, as discovered by analysts, is about 30 characters in length. So, we might impose a new characteristic, called **relative length**, (which is the length of the tweet divided by the average length), telling us the length of this tweet as compared to the average tweet. This tweet is actually 4.03 times longer than the average tweet, as shown:

$$\frac{121}{30} = 4.03$$

Picking out topics:

	this	wednesday	morn	are	this wednesday	?	Relative length
Word Count	1	1	1	1	1	1	4.03

	this	wednesday	morn	are	this wednesday	?	Relative length	Topic
Word Count	1	1	1	1	1	1	4.03	astronomy

Types of Data
Quantitative versus qualitative data:

- **Quantitative data:** This data can be described using numbers, and basic mathematical procedures, including addition, are possible on the set.
- **Qualitative data:** This data cannot be described using numbers and basic mathematics. This data is generally thought of as being described using "natural" categories and language.

- Example – coffee shop data - Say that we were processing observations of coffee shops in a major city using the following five descriptors (characteristics):
    - **Data: Coffee Shop**
    - Name of coffee shop
    - Revenue (in thousands of dollars)
    - Zip code
    - Average monthly customers
    - Country of coffee origin
- Each of these characteristics can be classified as either quantitative or qualitative, and that simple distinction can change everything. Let's take a look at each one:

- **Name of coffee shop** – Qualitative - The name of a coffee shop is not expressed as a number and we cannot perform math on the name of the shop.
- **Revenue** – Quantitative - How much money a cafe brings in can definitely be described using a number. Also, we can do basic operations such as adding up the revenue for 12 months to get a year's worth of revenue.
- **Zip code** – Qualitative - This one is tricky. A zip code is always represented using numbers, but what makes it qualitative is that it does not fit the second part of the definition of quantitative—we cannot perform basic mathematical operations on a zip code. If we add together two zip codes, it is a nonsensical measurement. We don't necessarily get a new zip code and we definitely don't get "double the zip code".
- **Average monthly customers** – Quantitative - Again, describing this factor using numbers and addition makes sense. Add up all of your monthly customers and you get your yearly customers.
- **Country of coffee origin** – Qualitative - We will assume this is a very small café with coffee from a single origin. This country is described using a name (Ethiopian, Colombian), and not numbers.

When trying to decide whether or not the data is qualitative or quantitative, ask yourself a few basic questions about the data characteristics:

- Can you describe it using numbers?

  No? It is **qualitative.**

  Yes? Move on to next question.
- Does it still makes sense after you add them together?

  No? They are **qualitative.**

  Yes? You probably have **quantitative** data.
- This method will help you classify most, if not all, data into one of these two categories.

- The difference between these two categories define the types of questions you may ask about each column. For a quantitative column, you may ask questions such as the following:
  - What is the average value?
  - Does this quantity increase or decrease over time (if time is a factor)?
  - Is there a threshold that if this number grew above or be too low would signal trouble for the company?
- For a qualitative column, none of the preceding questions can be answered; however, the following questions *only* apply to qualitative values:
  - Which value occurs the most and the least?
  - How many unique values are there?
  - What are these unique values?

Example

- **World alcohol consumption data:** The World Health Organization released a dataset describing the average drinking habits of people in countries across the world

```
import pandas as pd

# read in the CSV file from a URL
drinks = pd.read_csv('https://raw.githubusercontent.com/sinanuozdemir/
principles_of_data_science/master/data/chapter_2/drinks.csv')

# examine the data's first five rows
drinks.head()             # print the first 5 rows
```

	country	beer_servings	spirit_servings	wine_servings	total_litres_of_pure_alcohol	continent
0	Afghanistan	0	0	0	0.0	AS
1	Albania	89	132	54	4.9	EU
2	Algeria	25	0	14	0.7	AF
3	Andorra	245	138	312	12.4	EU
4	Angola	217	57	45	5.9	AF

- We have six different columns that we are working with in this example:
  - country: Qualitative
  - beer_servings: Quantitative
  - spirit_servings: Quantitative
  - wine_servings: Quantitative
  - total_litres_of_pure_alcohol: Quantitative
  - continent: Qualitative
- Let's look at the **qualitative column** continent. We can use Pandas in order to get some basic summary statistics about this non-numerical characteristic. The describe() method first identifies whether the column is likely quantitative or qualitative and then gives basic information about the column as a whole.
- drinks['continent'].describe()
- >> count 193
- >> unique 5
- >> top AF
- >> freq 53
- It reveals that the WHO has gathered data about five unique continents, the most frequent being AF (Africa), which occurred 53 times in the 193 observations.

- If we take a look at one of the quantitative columns and call the same method, we can see the difference in output, as shown:

  ```
  drinks['beer_servings'].describe()
  >> mean 106.160622
  >> min 0.000000
  >> max 376.000000
  ```

  Now we can look at the mean (average) beer serving per-person per-country (106.2 servings) as well as the lowest beer serving, zero, and the highest beer serving recorded, 376 (that's more than a beer a day)

Quantitative data

Quantitative data can be broken down, one step further, into *discrete* and *continuous* quantities. These can be defined as follows:

- **Discrete data**: This describes data that is counted. It can only take on certain values. Examples of discrete quantitative data include a dice roll, because it can only take on six values, and the number of customers in a café, because you can't have a *real* range of people.
- **Continuous data**: This describes data that is measured. It exists on an infinite range of values. A good example of continuous data would be a person's weight because it can be 150 pounds or 197.66 pounds (note the decimals). The height of a person or building is a continuous number because an infinite scale of decimals is possible. Other examples of continuous data would be time and temperature.

The FOUR levels of data

- A specific characteristic (feature/column) of structured data can be broken down into one of four levels of data. The levels are:
    - The nominal level
    - The ordinal level
    - The interval level
    - The ratio level

1. The Nominal Level

- Consists of data that is described purely by name or category
- Basic examples include gender, nationality, species, or yeast strain in a beer. They are not described by numbers and are therefore qualitative.
- We cannot perform mathematics on the nominal level of data except the basic *equality* and *set membership* functions, as shown in the following two examples:
  - *Being a tech entrepreneur* is the same as *being in the tech industry*, but not vice versa
  - A figure described as a square falls under the description of being a rectangle, but not vice versa

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- A **measure of center** is a number that describes what the data *tends to*. It is sometimes referred to as the *balance point* of the data. Common examples include the mean, median, and mode. In order to find the *center* of nominal data, we generally turn to the *mode* (the most common element) of the dataset. In WHO alcohol consumption data example the most common continent surveyed Africa is the center.
- Data at the nominal level is mostly categorical in nature. With only the mode as a basic measure of center, we are unable to draw conclusions about an *average* observation.

2. The Ordinal Level

- Data in the ***ordinal*** level provides us with a rank order, or the means to place one observation before the other; however, it does not provide us with relative differences between observations, meaning that while we may order the observations from first to last, we cannot add or subtract them to get any real meaning.
- Example: The *Likert* is among the most common ordinal level scales. Whenever you are given a survey asking you to rate your satisfaction on a scale from 1 to 10, you are providing data at the ordinal level. Your answer, which must fall between 1 and 10, can be ordered: eight is better than seven while three is worse than nine. However, differences between the numbers do not make much sense. The difference between a seven and a six might be different than the difference between a two and a one.

- **Mathematical operations allowed:** We inherit all mathematics from the nominal level (equality and set membership) and we can also add the following to the list of operations allowed in the ordinal level:
  - Ordering
  - Comparison
- **Ordering** refers to the natural order provided to us by the data.
- **As long as we are consistent in what defines the order, it does not matter what defines it.**
- **Example:** When speaking about the spectrum of visible light, we can refer to the names of colors—red, orange, yellow, green, blue, indigo, and violet. Naturally, as we move from left to right, the light is gaining energy and other properties. We may refer to this as a natural order. However, if needed, an artist may impose another order on the data, such as sorting the colors based on the cost of the material to make the said color

The ordinal level

- **Comparisons** are another new operation allowed at this level. At the nominal level, it would not make sense to say that one country was *naturally* better than another or that one part of speech is worse than another. At the ordinal level, we can make these comparisons. For example, we can talk about how putting a "7" on a survey is worse than putting a "10".
- **Measures of center:** the **median** is usually an appropriate way of defining the center of the data. The mean, however, would be impossible because division is not allowed at this level.

- Example: Imagine you have conducted a survey among your employees asking "how happy are you to be working here on a scale from 1-5", and your results are as follows: 5, 4, 3, 4, 5, 3, 2, 5, 3, 2, 1, 4, 5, 3, 4, 4, 5, 4, 2, 1, 4, 5, 4, 3, 2, 4, 4, 5, 4, 3, 2, 1
  - import numpy
  - results = [5, 4, 3, 4, 5, 3, 2, 5, 3, 2, 1, 4, 5, 3, 4, 4, 5, 4, 2, 1, 4, 5, 4, 3, 2, 4, 4, 5, 4, 3, 2, 1]
  - sorted_results = sorted(results)
  - print sorted_results
  - [1, 1, 1, 2, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3, 4, 4, 4, 4, 4, 4, 4, 4, 4,4, 4, 5, 5, 5, 5, 5, 5, 5]
  - print numpy.mean(results) # == 3.4375
  - print numpy.median(results) # == 4.0
- Turns out that the median is not only more sound, but makes the survey results look much better.

3. The Interval Level

- Data that can be expressed through very quantifiable means, and where much more complicated mathematical formulas are allowed. The basic difference between the ordinal level and the interval level is, well, just that—difference.
- Data at the interval level allows meaningful subtraction between data points.
- <u>Example:</u> Temperature is a great example of data at the interval level. If it is 100 degrees Fahrenheit in Texas and 80 degrees Fahrenheit in Istanbul, Turkey, then Texas is 20 degrees warmer than Istanbul.

- The operations allowed on the lower levels (ordering, comparisons, and so on), along with two other notable operations:
  - Addition
  - Subtraction
- **Measures of center**: At this level, we can use the median and mode to describe this data; however, usually the most accurate description of the center of data would be the **arithmetic mean**, more commonly referred to as, simply, "the mean".
- Only at the interval level and above that the arithmetic mean makes sense.

The interval level

- Example: Suppose we look at the temperature of a fridge containing a pharmaceutical company's new vaccine. We measure the temperature every hour with the following data points (in Fahrenheit): 31, 32, 32, 31, 28, 29, 31, 38, 32, 31, 30, 29, 30, 31, 26
  - import numpy
  - temps = [31, 32, 32, 31, 28, 29, 31, 38, 32, 31, 30, 29, 30, 31, 26]
  - print numpy.mean(temps) # == 30.73
  - print numpy.median(temps) # == 31.0
- the mean and median are quite close to each other and both are around 31 degrees.
- The question, *on average, how cold is the fridge?,* about 31, however the vaccine comes with a warning:
  - *Do not keep this vaccine at a temperature under 29 degrees*

- **Measures of variation**: in data science, it is also very important to mention how "spread out" the data is. The measures that describe this phenomenon are called **measures of variation**.
- A measure of variation (like the standard deviation) is a number that attempts to describe how spread out the data is. Along with a measure of center, a measure of variation can almost entirely describe a dataset with only two numbers.

- **Standard deviation:** SD is the most common measure of variation of data at the interval level and beyond. The standard deviation can be thought of as the "average distance a data point is at from the mean”
- The formula for standard deviation can be broken down into the following steps:
  - 1. Find the mean of the data.
  - 2. For each number in the dataset, subtract it from the mean and then square it.
  - 3. Find the average of each square difference.
  - 4. Take the square root of the number obtained in step three. This is the standard deviation.

The interval level

- For example, look back at the temperature dataset. Let's find the standard deviation of the dataset using Python:
  - import numpy
  - temps = [31, 32, 32, 31, 28, 29, 31, 38, 32, 31, 30, 29, 30, 31, 26]
  - mean = numpy.mean(temps) # == 30.73
  - squared_differences = []

    # empty list o squared differences
  - for temperature in temps:
  - difference = temperature - mean

    # how far is the point from the mean

- squared_difference = difference**2

  # square the difference
- squared_differences.append(squared_difference)

  # add it to our list
- average_squared_difference = numpy.mean(squared_differences)

  # This number is also called the "Variance"
- standard_deviation = numpy.sqrt(average_squared_difference)

  # We did it!
- print standard_deviation # == 2.5157

• All of this code led to us find out that the standard deviation of the dataset is around 2.5, meaning that "on average", a data point is 2.5 degrees off from the average temperature of around 31 degrees, meaning that the temperature could likely dip below 29 degrees again in the near future.

4. The Ratio Level

- After moving through three different levels with differing levels of allowed mathematical operations, the ratio level proves to be the strongest of the four.
- Not only can we define order and difference, the ratio level allows us to *multiply and divide* as well
- **Example1:** While Fahrenheit and Celsius are stuck in the interval level, the Kelvin scale of temperature boasts a natural zero. A measurement zero Kelvin literally means the absence of heat. It is a non-arbitrary starting zero. We can actually scientifically say that 200 Kelvin is twice as much heat as 100 Kelvin.

Many people may argue that Celsius and Fahrenheit also have a starting point (mainly because we can convert from Kelvin to either of the two). The real difference here might seem silly, but because the conversion to Celsius and Fahrenheit make the calculations go into the negative, it does not define a clear and "natural" zero.

- **Example2:** Money in the bank is at the ratio level. You can have "no money in the bank" and it makes sense that $200,000 is "twice as much as" $100,000.

The ratio level

- **Measures of center:** a new *type* of mean called the **geometric mean**. It is the square root of the product of all the values.
- Example, in our fridge temperature data, we can calculate the geometric mean.
  - import numpy
  - temps = [31, 32, 32, 31, 28, 29, 31, 38, 32, 31, 30, 29, 30, 31, 26]
  - num_items = len(temps)
  - product = 1.
  - for temperature in temps:
  - product *= temperature
  - geometric_mean = product**(1./num_items)
  - print geometric_mean # == 30.634

- Data at the ratio level is usually non-negative. many data scientists prefer the interval level to the ratio level. The reason for this restrictive property is because if we allowed negative values, the ratio might not always make sense.
- Consider that we allowed debt to occur in our money in the bank example. If we had a balance of \$50,000, the following ratio would not really make sense at all:

$$\frac{\$50,000}{-\$50,000} = -1$$

Recap

- At the nominal level, we deal with data usually described using vocabulary (but sometimes with numbers), with no order, and little use of mathematics.
- At the ordinal level, we have data that can be described with numbers and also have a "natural" order, allowing us to put one in front of the other.
- try to classify the following example as either ordinal or nominal.
  - The origin of the beans in your cup of coffee
  - The place someone receives after completing a foot race
  - The metal used to make the medal that they receive after placing in the said race
  - The telephone number of a client
  - How many cups of coffee you drink in a day

- Data is in the eye of the beholder
- Whenever you are faced with a new dataset, the first three questions you should ask about it are the following:
  - Is the data organized or unorganized?

    For example, does our data exist in a nice, clean row/column structure?
  - Is each column quantitative or qualitative?

    For example, are the values numbers, strings, or do they represent quantities?
  - At what level of data is each column?

    For example, are the values at the nominal, ordinal, interval, or ratio level?